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Roll No

BEPT-101(EX/ME)-GS

B.E./B.Tech., (Part Time) I Semester

Examination, June 2020

Grading System (GS)

Engineering Mathematics - I

Time : Three Hours

Maximum Marks : 70

- Note:** i) Attempt any five questions out of eight.
ii) All questions carry equal marks.

1. a) Obtain the Fourier series to represent

$$f(x) = \frac{1}{4}(\pi - x)^2, \quad 0 < x < 2\pi$$

- b) Express $f(x) = |x|$, $-\pi < x < \pi$ as a Fourier series.

2. a) Find Laplace transform of the followings.

i) $t^3 e^{-3t}$ ii) $\frac{\sin 2t}{t}$

- b) Find inverse Laplace transform of the following functions.

i) $\frac{5s + 3}{(s - 1)(s^2 + 2s + 5)}$

ii) $\log\left(\frac{s+1}{s-1}\right)$

3. a) Using Laplace transform, solve the equation

$$y'' - 3y' + 2y = 4t + e^{3t}$$

when $y(0) = 1$ and $y'(0) = -1$

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b) Apply Convolution theorem to evaluate $L^{-1} \left\{ \frac{s}{(s^2 + a^2)^2} \right\}$

4. a) Solve the following differential equation

$$x \frac{d^2 y}{dx^2} + (1-x) \frac{dy}{dx} - y = e^x$$

Given that $y = e^x$ is a part of complementary function.

- b) Using the method of removal of first derivative, solve the differential equation

$$x^2 \frac{d^2 y}{dx^2} - 2(x^2 + x) \frac{dy}{dx} + (x^2 + 2x + 2)y = 0$$

5. a) Solve by Power series method $\frac{d^2 y}{dx^2} + y = 0$

b) Solve $x^2 p + y^2 q = nxy$.

6. a) Solve $(y + z)p + (z + x)q = x + y$

b) Solve $z = px + qy + pq$

7. a) Use Charpit method, to solve

$$px + qy = pq$$

b) Solve $(D - D' - 1)(D - D' - 2)z = e^{2x-y}$

8. a) Find grad ϕ , when ϕ is given by

$$\phi = 3x^2 y - y^3 z^2 \text{ at the point } (1, -2, -1)$$

- b) Find divergence and curl of the vector

$$\vec{v} = (xyz)\hat{i} + (3x^2 y)\hat{j} + (xz^2 - y^2 z)\hat{k}$$

at the point $(2, -1, 1)$

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